Dorado: Two-Stage Reasoning Post-Training with Non-Verifiable Data Cold-Start and Dual-Reward Refinement

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Abstract

Existing post-training paradigms, such as RL-Zero and Long-CoT SFT, have shown promising improvements in the reasoning capabilities of LLMs. However, these gains often require substantial computational resources; the former can produce reasoning trajectories misaligned with human understanding, while the latter tends to generate overly verbose outputs, together compromising the user experience. To address these limitations, we introduce Dorado, a lightweight two-stage training framework that enhances reasoning capabilities while improving chat ability. It first leverages non-verifiable instruction data to cold-start for reasoning, followed by a dual-reward DPO stage that combines verifiable correctness rewards with non-verifiable preference rewards for further refinement. Compared to the strong baseline SimpleRL-Zero, Dorado achieves an average 2.3% higher Pass@1 accuracy across five widely used mathematical benchmarks, while also exhibiting approximately 20% improvement on the general chat benchmark AlpacaEval 2.0. Analysis also shows that Dorado cuts training cost by up to 72× and has better token efficiency.

1. Introduction

Advanced post-training methodologies for large language models (LLMs) have become increasingly important to unlock reasoning potential and ensure alignment after establishing fundamental capabilities from pre-training (Jaech et al., 2024; Guo et al., 2025). Particularly, reinforcement learning (RL) (Gandhi et al., 2025; Zeng et al., 2025; Tu et al., 2025b) and supervised fine-tuning (SFT) on extended Chain-of-Thought (CoT) data (Muennighoff et al., 2025; Ye et al., 2025; Fu et al., 2025) have proven instrumental in unlocking the potential of base models for reasoning tasks.

Current online RL methods remain computationally prohibitive. For example, SimpleRL (Zeng et al., 2025) with a 7B model requires approximately 1.5 days on 32 H100 GPUs, while PURE (Cheng et al., 2025) necessitates a similarly heavy setup. To address this computational bottleneck, recent works have adopted offline maximum likelihood estimation (MLE) objectives as more efficient alternatives, including DPO (Rafailov et al., 2023), KTO (Ethayarajh et al., 2024), and IPO (Azar et al., 2024). By modeling rewards implicitly, these approaches bypass the complexity of explicit value function estimation while theoretically approximating off-policy online RL (Pang et al., 2024; Swamy et al., 2025). However, a critical limitation persists: similar to their online RL counterparts, these offline strategies primarily optimize for reasoning correctness, without explicitly encouraging clarity or coherence in reasoning traces or conversational behavior. As a result, when applied directly to base models, they often do not lead to clearer or more interpretable reasoning and offer limited improvements in general conversational capabilities. As shown in Figure 1, existing strong methods achieve near-zero win rates on the general chat benchmark AlpacaEval 2.0 (Dubois et al., 2025).

Parallel to these optimization-centric approaches, a prevalent alternative involves directly applying SFT using long CoT datasets (Ye et al., 2025; Hugging Face, 2025). While intended to elicit latent reasoning capabilities, this strategy introduces drawbacks. First, it relies on inducing excessive verbosity to maximize accuracy, which increases inference cost and produces outputs that are tedious for human users to parse (Yu et al., 2025). Second, its effectiveness is scale-dependent. As noted by Ye et al. (2025), while long CoT patterns can benefit models such as 32B models, they often fail to yield consistent improvements for smaller models such as 7B models, limiting their applicability to low-resource scenarios. Achieving strong long-CoT reasoning performance also requires costly training by distilling detailed reasoning trajectories from large-scale teacher models (DeepSeek-AI, 2025).

On the other hand, aligning reasoning trajectories with human understanding has become an increasingly important

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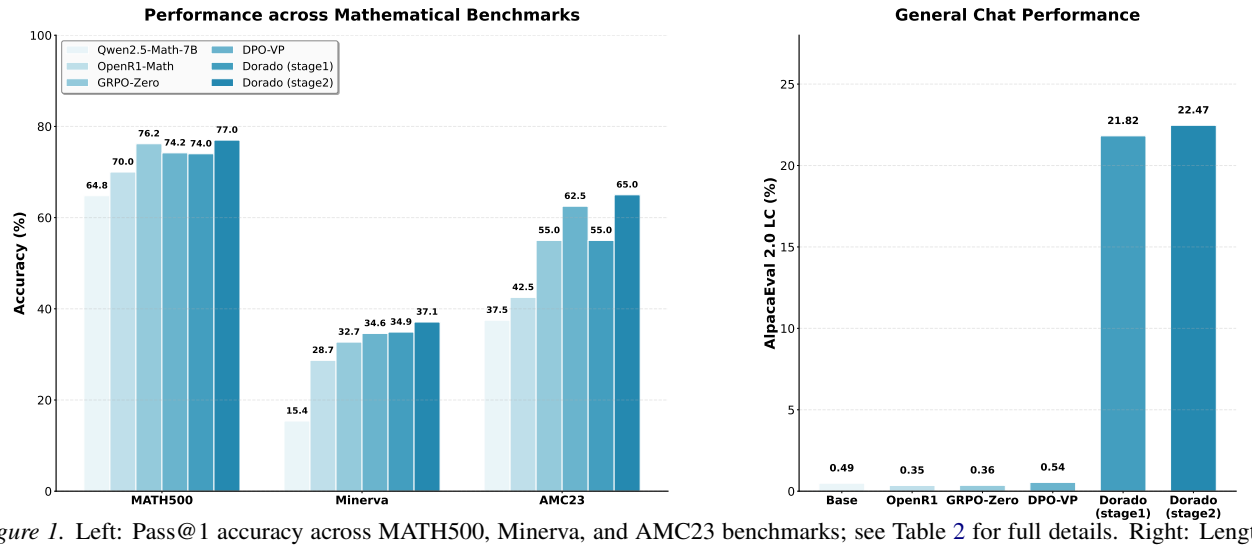

Figure 1. Left: Pass@1 accuracy across MATH500, Minerva, and AMC23 benchmarks; see Table 2 for full details. Right: Length-controlled Win Rates (LC) of model responses against GPT-4 Turbo on the general chat benchmark AlpacaEval 2.0.

consideration in reasoning post-training (Bhaskar et al., 2025). Widely used post-training pipelines, such as those adopted in Qwen2.5-Math-Instruct (Yang et al., 2024), rely heavily on large-scale SFT over verifiable math data to teach solution patterns, followed by GRPO (Shao et al., 2024) to further unlock reasoning performance. While effective, this post-training pipeline is computationally expensive and nevertheless widely adopted in current practice. Intuitively, employing verifiable data (e.g., math) to cold-start a base model pre-trained on massive math corpora appears to be a logical standard. However, we propose taking a step back to re-evaluate the fundamental deficiencies of these base models. Assuming that a model has been exposed to massive math tokens during pre-training, it is plausible to hypothesize that it may already encode the substantial problem-solving logic within its latent space. Consequently, the primary deficit may not be a lack of domain knowledge, but rather the absence of conversational alignment required to articulate reasoning clearly. Despite this, existing approaches persist in relying on costly domain-specific math SFT to explicitly guide the model toward target solutions.

To empirically validate this hypothesis, we propose **Dorado**, our main method contribution for aligning reasoning articulation with *hybrid supervision signals*, combining non-verifiable instruction data with dual math reward feedback. Dorado adopts a simple two-stage design. We first cold-start the base model using non-verifiable general chat instructions, which encourage clear and human-aligned expression of latent reasoning without imposing rigid solution templates (Stage 1). Inspired by prior findings that offline verifiable rewards can effectively optimize solution correctness (Tu et al., 2025a), we further introduce an offline *dual-reward optimization mechanism* on math data (Stage 2). Specifically, a rule-based reward enforces solution correctness on verifiable tasks, while a learned reward model prioritizes response quality for non-verifiable objectives. By combining these complementary signals, Dorado reinforces correctness without sacrificing articulation quality, enabling efficient post-training without costly additional online reinforcement learning.

As shown in Figure 1, a key empirical finding emerges in Stage 1 of Dorado: despite being trained exclusively on non-verifiable and fully subjective data, the model not only achieves improved conversational alignment, yielding an approximately 20% gain in length-controlled performance on AlpacaEval 2.0 (Dubois et al., 2025), but also demonstrates enhanced reasoning capability, surpassing the GRPO-Zero baseline (Shao et al., 2024). Notably, this stage also leads to a modest improvement in general chat generation ability, suggesting a positive interaction between reasoning competence and conversational generation. In Stage 2, the model further strengthens its mathematical reasoning ability and ultimately achieves the best performance among a range of recent strong baselines.

We summarize our main contributions as follows:

- We present the first study on leveraging non-verifiable general chat data to post-train a base model specialized for verifiable reasoning tasks, revealing that reasoning competence and conversational generation are not isolated and can be jointly enhanced under this setting.

- We propose Dorado, an offline post-training framework that combines a lightweight chat-oriented cold start with offline dual reward optimization on mathematical prompts, enabling effective reasoning enhancement without relying on long CoT SFT or online RL.

- Dorado achieves strong performance relative to recent baselines while being highly efficient, reducing training budget by up to 72× compared to prominent online RL and 5× compared to offline baselines, and delivering superior token efficiency.
- Dorado can be even seamlessly integrated with existing RL-trained models to further and continually enhance their reasoning performance, demonstrating strong generalization and compatibility.

2. Dorado

2.1. General-Chat Boosting on Verifiable Base Models

Diverging from prevailing paradigms that cold-start math-pretrained models via SFT on long CoT mathematical data, we prioritize equipping the base model π_0 with robust general capabilities. Specifically, we cold-start the model using a curated dataset of high-quality non-verifiable instruction data, inspired by Yin et al. (2025) which demonstrated enhanced alignment in a data-efficient manner. Adopting a similar construction methodology, our dataset encompasses diverse open-ended tasks where ground-truth answers cannot be directly verified through automated rules, spanning domains such as creative writing. Formally, this cold-start phase employs standard SFT:

$$\mathcal{L}_{\text{GC-Boost}}(\pi_G) = \mathbb{E}_{(Q,r)\sim\mathcal{D}_{\text{chat}}}\left[-\log \pi_0(r \mid Q)\right] \quad (1)$$

where Q represents the instruction prompt, r denotes the corresponding response, and $\mathcal{D}_{\text{chat}}$ refers to the general instruction dataset. This initialization establishes a critical foundation of conversational alignment before the model undergoes specialization in mathematical reasoning.

2.2. Offline DPO Training with Dual Rewards

Self-Generated Math Responses We adopt a self-generation strategy to construct training responses for math tasks. Specifically, for each mathematical question Q_k, the cold-started model π_G generates multiple candidate trajectory responses $\mathcal{R}_k = \{r_{1,k}, r_{2,k}, \ldots, r_{n,k}\}$. Each response $r_{i,k}$ represents a complete solution trajectory, comprising intermediate reasoning steps and the final answer.

Verifiable Reward For mathematical problems possessing verifiable ground-truth answers, we employ deterministic rule-based verifiers to automatically assess correctness (e.g., parsing the final answer enclosed within the `\boxed{}` format). Given a response $r_{i,k}$ to question Q_k with ground-truth answer A_k, the verifier extracts the predicted answer $\hat{A}_{i,k}$ and assigns a binary correctness label:

$$\ell_{\text{Rule}}(r_{i,k}) = \begin{cases} \text{correct} & \text{if } \hat{A}_{i,k} = A_k \\ \text{incorrect} & \text{otherwise} \end{cases} \quad (2)$$

We first apply this rule-based verification to all generated candidate responses. We filter out questions where all candidate responses are incorrect, as they lack a positive reference. For questions that yield a mix of correct and incorrect responses, we construct the training pair (r_k^+, r_k^-) by sampling one correct response as the chosen r_k^+ and one incorrect response as the rejected r_k^-. Questions containing exclusively correct responses are then forwarded to the subsequent Non-verifiable Reward stage for fine-grained ranking.

Non-verifiable Reward Building upon the General-Chat Boosting phase, which has equipped the base model with robust conversational capabilities, we further introduce a reward model to evaluate response quality beyond mere correctness. While rule-based verifiers act as a hard constraint for accuracy, we utilize a reward model to assess nuanced attributes such as clarity, logical coherence, and alignment with human preferences. The reward model assigns a scalar score $s_{\text{RM}}(r_{i,k})$ that reflects the overall quality of the reasoning trajectory. This is particularly crucial for distinguishing among multiple correct solutions. Unlike standard RLVR, which typically treats all correct trajectories equally and ignores the gradient variance between them, our approach incentivizes the model to produce reasoning that is not only accurate but also articulate and pedagogically valuable.

For each question Q_k, we specifically focus on the subset of candidate responses deemed correct by the rule-based verifiers, denoted as $\mathcal{R}_k^{\text{correct}}$. To align the model towards higher-quality reasoning, we construct preference pairs entirely within this correct subset by selecting the highest-scored response as the chosen response r_k^+ and the lowest-scored response as the rejected response r_k^-:

$$r_k^+ = \arg\max_{r\in\mathcal{R}_k^{\text{correct}}} s_{\text{RM}}(r), \quad r_k^- = \arg\min_{r\in\mathcal{R}_k^{\text{correct}}} s_{\text{RM}}(r) \quad (3)$$

This strategy ensures that the preferred responses exhibit both correctness and superior quality, while dispreferred responses, although factually correct, may be poorly articulated or unnecessarily complex.

Offline Optimization via DPO Finally, we perform offline optimization utilizing the constructed preference pairs via the DPO algorithm. The policy π_θ is optimized to minimize the standard DPO loss:

$$\mathcal{L}_{\text{DPO}}(\pi_\theta) = -\,\mathbb{E}_{(Q_k, r_k^+, r_k^-)}\left[\log\sigma\left(\beta\left(\log\frac{\pi_\theta(r_k^+ \mid Q_k)}{\pi_G(r_k^+ \mid Q_k)} - \log\frac{\pi_\theta(r_k^- \mid Q_k)}{\pi_G(r_k^- \mid Q_k)}\right)\right)\right] \quad (4)$$

where σ denotes the sigmoid function, β is the hyperparameter controlling the deviation from the reference policy π_G,

Table 1. Comparison of training data size and computational resources across different models. With the exception of Dorado, baseline statistics are derived from Tu et al. (2025a). The training workflow for Dorado (≈4 hours total) consists of approximately 1 hour for GC-Boosting SFT, 1–1.5 hours for offline response generation and reward labeling, and 1–1.5 hours for DPO training.

	Qwen2.5-Math-7B-Instruct	**Eurus-2-7B-PRIME**	**Qwen2.5-7B-SimpleRL-Zero**	**Qwen2.5-7B-DPO-VP**	**Dorado (Ours)**
Base Model	Qwen2.5-Math-7B	Qwen2.5-Math-7B	Qwen2.5-Math-7B	Qwen2.5-Math-7B	Qwen2.5-Math-7B
SFT Data	2.5M (open-source and in-house)	230K	0	0	10K GC
RM Data	618K (in-house)	0	0	0	0
RM	Qwen2.5-Math-RM (72B)	Eurus-2-7B-SFT	None	None	Skywork-Reward-8B
Self-improve Method	RL + ORM	RL + PRM	RL + VR	DPO + VR	DPO + DR
Self-improve Data	66K	150K	8K MATH	8K MATH	8K MATH
GPUs & Training Times	Unknown but likely massive	8 A100 for few days	32 H100 for 1.5 days	4 A800 for 20 hours	**4 PRO6000 for 4 hours (96G each)**

and the optimization objective effectively guides the model to prioritize responses that are simultaneously mathematically precise and pedagogically articulate.

3. Experiments

3.1. Backbone Models and Implementation Details

In this section, we systematically investigate the impact of our framework on varying base models, including Qwen2.5-Math-1.5B, Qwen2.5-Math-7B (Qwen et al., 2025), and Qwen2.5-3B-Base (Qwen et al., 2025). For the General-Chat Boosting stage, we adopt a method similar to Yin et al. (2025) by generating 10,000 high-quality open-ended instruction-following pairs using GPT-4o, upon which we SFT the models for 3 epochs with a learning rate of 2×10^{-5}. Subsequently, for the offline DPO phase, we employ verifiable and non-verifiable reward filtering on self-generated responses (n = 5), setting the sampling temperature $T = 1.0$ to ensure diversity. We perform DPO training for one epoch with $\beta = 0.01$ and a learning rate of 1×10^{-6}. To align with established RL baselines (Zeng et al., 2025; Cheng et al., 2025), we utilize 8,000 problems from the MATH dataset (Difficulty Levels 3–5) as self-improvement prompts.

3.2. Baselines and Evaluation Benchmarks

In our experiments, we primarily utilize Qwen2.5-Math-7B as base backbone model and compare our method against the following strong baselines across four categories: (1) **Official Models:** We include **Qwen2.5-Math-7B-Instruct** (Qwen et al., 2025), the official instruction-tuned variant known for its strong mathematical capabilities; (2) **Long-CoT Baselines:** We compare against models trained on specialized long chain-of-thought data, specifically **LIMO** (Ye et al., 2025), **s1** (Muennighoff et al., 2025), and a variant fine-tuned on a random 10k subset of **OpenR1-Math** (Hugging Face, 2025); (3) **Online RL Baselines:** We select leading online reinforcement learning methods, including **rStar-Math** (Guan et al., 2025), which employs self-evolution MCTS with a process preference model; **Eurus-2-PRIME** (Cui et al., 2025), trained via process RL with implicit rewards; **GRPO-Zero** (Shao et al., 2024), trained by GRPO with verifiable rewards without extra SFT data; **Simple-RL-Zero/Zoo** (Zeng et al., 2025), trained by PPO with verifiable rewards (Zero and Zoo refer to two versions released at different time points); and **PURE-VR** (Cheng et al., 2025), which utilizes verifiable rewards with min-form credit assignment; and (4) **Offline RL Baselines:** We compare against iterative offline approaches such as **DPO-R1-Zero** (Zhang et al., 2025) and **DPO-VP** (Tu et al., 2025a), which apply iterative DPO without extra SFT data. For mathematical evaluation, we assess performance across diverse benchmarks, including MATH500, Minerva, AMC 2023, AIME 2024 (Zhang & Math-AI, 2024), AIME 2025 (Zhang & Math-AI, 2025), and OlympiadBench (He et al., 2024). To evaluate general chat capabilities, we utilize the widely-used AlpacaEval 2.0 (Dubois et al., 2025).

4. Main Results

4.1. Training Budget Analysis

As presented in Table 1, we conduct a comparative analysis of computational efficiency and data usage across different frameworks. The results highlight the extreme resource efficiency of Dorado. While the official Qwen2.5-Math-7B-Instruct relies on millions of SFT samples and massive computing resources, and process-based methods like Eurus-2-7B-PRIME require training on 8 A100 GPUs for several days with hundreds of thousands of samples, Dorado achieves competitive performance with lowest cost. Specifically, compared to the leading online RL baseline Qwen2.5-7B-SimpleRL-Zero, which demands a massive

Table 2. Pass@1 accuracy on mathematical benchmarks. We report the accuracy (%) across varying datasets. All models are based on Qwen2.5-Math-7B; model names are shortened by removing this common prefix. Bold indicates the best result, and underline indicates the second best. Results marked with * are reproduced by us, while others are sourced from Tu et al. (2025a).

Model Variation	MATH500	Minerva	Olympiad	AMC23	AIME24	Avg.
Official Models						
Base	64.8	15.4	25.6	37.5	16.7	32.0
Instruct	**83.2**	33.5	38.4	62.5	20.0	47.5
Long CoT Baselines						
LIMO	70.2	21.0	37.0	45.0	16.7	38.0
S1	72.0	32.4	37.5	55.0	13.3	42.0
OpenR1-Math*	70.0	28.7	32.3	42.5	10.0	36.7
Online RL Baselines						
rStar-Math	78.4	-	**47.1**	47.5	26.7	-
Eurus-2-PRIME	74.0	**39.7**	35.6	57.5	23.3	46.0
GRPO-Zero	76.2	32.7	38.1	55.0	16.7	43.7
Simple-RL-Zero	78.0	33.1	36.6	60.0	26.7	46.9
Simple-RL-Zoo	80.4	**39.7**	38.8	57.5	26.7	48.7
PURE-VR	79.8	36.8	41.9	60.0	20.0	47.7
Offline RL Baselines						
DPO-R1-Zero	76.8	30.9	37.9	62.5	26.7	47.0
DPO-VP*	74.2	34.6	38.1	62.5	23.3	46.5
Ours (Dorado Series)						
GC-Boost (Stage 1)	74.0	34.9	37.6	55.0	23.3	45.0
+ Offline DR (Stage 2)	77.0	37.1	36.9	**65.0**	**30.0**	**49.2**

Table 3. Pass@k ($k \in \{1, 8, 32\}$) results on AIME 2025 and AMC 2023 benchmarks. We highlight the **best** performance in bold.

Method	AIME25			AMC'23		
	Pass@1	Pass@8	Pass@32	Pass@1	Pass@8	Pass@32
Qwen2.5-Math-7B	3.3	23.3	33.3	37.5	75.0	89.0
OpenR1	**20.0**	23.3	26.7	42.5	60.0	80.0
SimpleRL-Zero	**20.0**	26.7	33.3	50.0	72.5	85.0
Dorado (Stage 1)	16.7	**30.0**	36.7	55.0	77.5	82.5
Dorado (Stage 2)	16.7	**30.0**	**46.7**	**65.0**	**90.0**	**95.0**

budget of 32 H100 GPUs for 1.5 days (approximately 1,152 GPU hours), Dorado completes training in just 4 hours using 4 PRO6000 GPUs (16 GPU hours), representing a remarkable 72× reduction in computational overhead. Furthermore, even when compared to the efficient offline baseline Qwen2.5-7B-DPO-VP, which utilizes 4 A800 GPUs for 20 hours, Dorado achieves more than 5× speedup.

4.2. Main Results on Math Benchmarks

As shown in Table 2, we compare Dorado against baselines on five mathematical benchmarks. Overall, Dorado achieves the best average accuracy of 49.2%, surpassing the official Qwen2.5-Math-7B-Instruct model (47.5%) as well as strong online RL baselines such as Simple-RL-Zoo (48.7%). Notably, Dorado demonstrates consistent improvements across datasets, with particularly strong gains on more challenging benchmarks such as AMC'23 and AIME'24. Even without the offline Dual-Reward (Stage 2) optimization, the cold-start GC-Boost (Stage 1) delivers competitive performance: by fine-tuning the base model solely with non-verifiable data, it reaches an average accuracy of 45.0%, outperforming more complex baselines including GRPO-Zero (43.7%) and OpenR1-Math (36.7%), while requiring significantly fewer data samples and computing hours. Building on this foundation, the Stage 2 further improves performance across multiple benchmarks, with the largest margin observed on AIME'24, where Dorado achieves 30.0% accuracy, establishing a clear lead over all evaluated methods.

4.3. Evaluation on Pass@k

To assess robustness and exploration capability, we evaluate Pass@k on AIME'25 and AMC'23. As shown in Table 3, both Long-CoT SFT and RL-Zero approaches exhibit a trade-off between Pass@1 and Pass@k. For example, OpenR1 improves Pass@1 on AIME'25 from 3.3% to 20.0% but reduces Pass@32 from 33.3% to 26.7%, and

Table 4. Average token count of model responses across five mathematical benchmarks (MATH500, Minerva, Olympiad, AMC'23, and AIME'24). Full results are reported in Table 10.

Model	Avg. Tokens
Qwen2.5-Math-7B	**667**
Qwen2.5-Math-7B-Instruct	850
Eurus-2-7B-PRIME	912
Qwen2.5-7B-Simple-RL	777
Qwen2.5-7B-PURE-VR	807
Qwen2.5-7B-DPO-VP	881
Qwen2.5-Math-7B-OpenR1	3185
Dorado	736

shows a similar regression on AMC'23, where Pass@32 drops from 89.0% to 80.0%. Likewise, SimpleRL-Zero increases Pass@1 yet yields only limited gains in Pass@k, with Pass@32 remaining at 33.3% on AIME'25 and decreasing to 85.0% on AMC'23, indicating concentration on a narrow set of solution trajectories. In contrast, Dorado's improvements in Pass@k primarily emerge in Stage 2. Stage 1 mainly improves conversational alignment and Pass@1, whereas Stage 2 substantially expands reasoning diversity as reflected by higher Pass@k. On AMC'23, Dorado Stage 2 achieves a Pass@1 of 65.0% while boosting Pass@32 to 95.0%. On the more challenging AIME'25 benchmark, Stage 2 raises Pass@32 to 46.7%, significantly exceeding both the base model and prior baselines.

4.4. Token Efficiency Analysis

Beyond accuracy, we evaluate the inference efficiency of different models by analyzing the average generated token counts across benchmarks, as detailed in Table 4. A notable trend in recent Long-CoT approaches, exemplified by Qwen2.5-Math-7B-OpenR1, is a drastic increase in computational cost. This model generates an average of 3,185 tokens, which is nearly four times the output length of standard models. In contrast, Dorado demonstrates exceptional conciseness with an average of only 736 tokens. This proves significantly more efficient than the official Instruct model (850 tokens) and other RL baselines such as Eurus-2-7B-PRIME (912 tokens) and Qwen2.5-7B-DPO-VP (881 tokens). These findings suggest that Dorado encourages the model to internalize precise reasoning patterns rather than relying on redundant or excessively verbose chains of thought, thereby achieving a superior balance between computational efficiency and mathematical accuracy.

4.5. Generalizability of Dorado

To evaluate the generalizability of Dorado, we conduct experiments across multiple parameter scales, model architectures, and initialization baselines. As shown in Table 5,

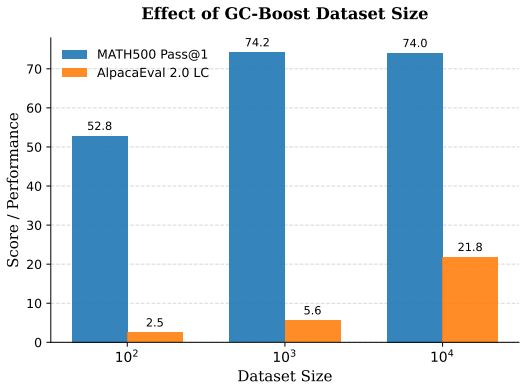

Figure 2. Ablation of Dorado Stage 1 with varying amounts of general chat data, showing the impact on both reasoning performance and general chat ability.

Dorado consistently improves performance across all evaluated settings. On the smaller Qwen2.5-Math-1.5B backbone, the GC-Boost stage improves the average accuracy from 23.7% to 37.8%, and further incorporating the offline dual-reward optimization increases performance to 39.9%, demonstrating that the gains persist at smaller model scales. On Qwen2.5-3B, GC-Boost substantially improves the average accuracy from 22.3% to 30.7%, and applying the offline dual-reward stage further increases performance to 31.4%, which is on par with GRPO at 31.1%. Dorado is also complementary to existing reinforcement learning refinements: when applied to the strong RL-aligned baseline Qwen2.5-Math-7B-SimpleRL-Zero, GC-Boost improves the average accuracy from 46.9% to 49.4%. Incorporating the offline dual-reward stage results in avg accuracy of 49.5% in this setting. We hypothesize that the marginal gain from the offline dual-reward stage reflects overlap with the existing RL alignment in SimpleRL-Zero, which already incorporates similar math optimization signals, leaving limited room for additional improvements. In addition, consistent improvements are observed when applying the GC-Boost stage to Qwen2.5-Math-7B-OpenR1.

4.6. Effect of Stage 1 Data Scale

To further explore the impact of general chat data scale in Stage 1, we vary the size of the GC-Boost dataset and evaluate both reasoning performance and general chat ability, as shown in Figure 2. Increasing the amount of general chat data leads to a clear and consistent improvement in general chat performance on AlpacaEval 2.0. In contrast, the gains in reasoning performance are not strictly monotonic and tend to saturate as the dataset size increases. This observation suggests that while scaling general chat data is effective for enhancing conversational ability, its effect on reasoning performance is more nuanced. We leave a deeper investigation of the underlying mechanisms and optimal scaling

Table 5. Generalization ability. We consider two pre-trained base models (*i.e.*, Qwen2.5-Math-1.5B, Qwen2.5-3B) and two post-trained reasoning models (*i.e.*, Qwen2.5-Math-7B-SimpleRL-Zero, Qwen2.5-Math-7B-OpenR1) as the backbone. We also provide an analysis of general chat ability in Appendix A.1.

Model	MATH	Minerva	Olympiad	AMC	AIME24	Avg.
Based on Qwen2.5-Math-1.5B						
Base	30.0	11.4	22.1	35.0	**20.0**	23.7
OpenR1	46.8	14.3	16.0	27.5	6.7	22.3
GRPO	70.2	19.1	30.5	52.5	13.3	37.1
GC-Boost	**71.0**	31.3	**32.3**	47.5	6.7	37.8
+ Offline DR	70.4	**32.7**	31.4	**55.0**	10.0	**39.9**
Based on Qwen2.5-3B						
Base	48.2	14.0	20.2	22.5	6.7	22.3
OpenR1	45.4	13.2	16.9	17.5	0.0	18.6
GRPO	**63.8**	23.5	**26.4**	**35.0**	6.7	31.1
GC-Boost	61.4	26.8	25.9	32.5	6.7	30.7
+ Offline DR	62.6	**27.9**	26.2	30.0	**10.0**	**31.4**
Based on Qwen2.5-Math-7B-SimpleRL-Zero						
Base	78.0	33.1	36.6	60.0	**26.7**	46.9
GC-Boost	79.2	39.3	40.1	**65.0**	23.3	49.4
+ Offline DR	**79.8**	**40.4**	**41.3**	62.5	23.3	**49.5**
Based on Qwen2.5-Math-7B-OpenR1						
Base	70.0	28.7	32.3	42.5	10.0	36.7
GC-Boost	**74.4**	**31.6**	**38.1**	**55.0**	**16.6**	**43.1**

Table 6. Ablation study of Dorado Stage 2. VR and NVR denote verifiable and non-verifiable rewards, respectively.

Variant	MATH500	Minerva	Olympiad	AMC23	AIME24	Avg.
GC-Boost (Stage 1)	74.0	34.9	37.6	55.0	23.3	45.0
+ Offline DR (Stage 2)	77.0	**37.1**	36.9	**65.0**	**30.0**	**49.2**
Stage 1 + GRPO	77.6	**37.1**	**39.7**	55.0	23.3	46.5
Stage 1 + Offline VR	76.4	35.7	38.4	57.5	20.0	45.6
Stage 1 + Offline NVR	**78.2**	35.3	38.7	52.2	20.0	44.9

strategies for future work.

4.7. Ablation Study of Dorado Stage 2

Table 6 presents an ablation study of the Stage 2 design in Dorado. Starting from the GC-Boost model (Stage 1), introducing the offline dual-reward mechanism substantially improves the average accuracy from 45.0% to 49.2%, while replacing Stage 2 with GRPO or using only a single reward signal, either verifiable or non-verifiable, leads to noticeably weaker performance. We hypothesize that the two reward signals play complementary roles in the offline setting, where the verifiable reward enforces correctness and the non-verifiable reward encourages higher-quality and more diverse reasoning behaviors.

4.8. Qualitative Analysis

Table 7 provides a qualitative comparison of an algebraic word problem. While the baseline Qwen2.5-Math-7B reaches the correct answer, it relies on unnecessary external tool calls to verify simple arithmetic (e.g., computing $18 + 12$), resulting in fragmented reasoning. In contrast, Dorado produces a coherent, purely textual solution without explicitly using additional mathematics-specific supervised fine-tuning. By suppressing superfluous tool usage, Dorado delivers reasoning that is not only correct but also more concise and easier to interpret.

5. Related Works

Resource-Efficient Post-Training for Reasoning. The release of DeepSeek-R1 (Guo et al., 2025) has catalyzed a paradigm shift towards efficient post-training strategies. While data-centric approaches like LIMO (Ye et al., 2025) and S1 (Muennighoff et al., 2025) highlight the value of curated long-chain data, and online RL methods such as SimpleRL-Zero (Zeng et al., 2025), PURE (Cheng et al., 2025), and Logic-RL (Xie et al., 2025) have refined credit

Table 7. Case Analysis. Qwen2.5-Math-7B (Left) includes redundant code verification, while Dorado (Right) provides a more concise, human-aligned mathematical derivation. We also provide a detailed analysis of reasoning response quality in Appendix A.2.

Question: Cities A and B are 45 miles apart. Alicia lives in A and Beth lives in B. Alicia bikes towards B at 18 miles per hour. Leaving at the same time, Beth bikes toward A at 12 miles per hour. How many miles from City A will they be when they meet?

Qwen2.5-Math-7B:

Let's solve the problem step-by-step.

1. **Understand the Problem:** Alicia and Beth are 45 miles apart. Alicia (18 mph) and Beth (12 mph) move towards each other.
2. **Calculate the Relative Speed:** $18 + 12 = 30$ miles per hour.
3. **Calculate the Time Until They Meet:** $t = \frac{45}{30} = 1.5$ hours.
4. **Calculate the Distance Alicia Travels:** $18 \times 1.5 = 27$ miles.

So, they will meet 27 miles from City A. Let's confirm this with Python code.

```
# Given values
distance = 45; alicia_v = 18; beth_v = 12
# Relative speed & time
rel_speed = alicia_v + beth_v
time = distance / rel_speed
# Distance Alicia travels
print(alicia_v * time)
```

Output: 27.0

The calculation confirms that Alicia and Beth will meet 27 miles from City A. Therefore, the final answer is $\boxed{27}$.

Qwen2.5-Math-7B-Dorado:

To determine how many miles from City A Alicia and Beth will meet, we can use the concept of relative speed.

1. **Calculate combined speed:**

$$\text{Speed} = 18 + 12 = 30 \text{ mph}$$

2. **Determine time to meet:**

$$\text{Time} = \frac{45 \text{ miles}}{30 \text{ mph}} = 1.5 \text{ hours}$$

3. **Calculate distance from City A:**

$$\text{Distance} = 18 \text{ mph} \times 1.5 \text{ hours} = 27 \text{ miles}$$

Thus, Alicia and Beth will meet 27 miles from City A.

The final answer is: $\boxed{27}$

assignment and logical consistency, these pipelines often remain computationally demanding. Although recent offline initiatives like DPO-VP (Tu et al., 2025a) have begun to explore self-improvement under constrained budgets, our work establishes a more streamlined alternative by exploiting the synergy between *verifiable* and *non-verifiable* rewards. We demonstrate that a generative chat cold start (GC-Boost)—relying merely on coarse-grained selection—is sufficient to prime strong reasoning capabilities without expensive online search. Furthermore, by integrating our proposed Dual Reward (DR) mechanism, the framework achieves continual performance gains, offering a significantly more resource-efficient pathway than the widely adopted SFT-Math+RL pipeline while delivering state-of-the-art results. Meanwhile, we note concurrent work such as RLMT (Bhaskar et al., 2025), which also explores integrating multiple supervision signals. However, RLMT primarily focuses on CoT–based supervision to enhance generative chat ability, whereas Dorado follows a fundamentally different design paradigm.

Preference Optimization via DPO. Recent advancements have successfully adapted DPO to the reasoning domain, moving beyond standard supervised fine-tuning (Xu et al., 2024; Zhang et al., 2025). Significant progress has been made in establishing rigorous theoretical frameworks for iterative DPO (Xiong et al., 2024) and extending these principles to multi-turn, tool-integrated (Xiong et al., 2025), and length-regularized scenarios (Liu et al., 2024). Concurrently, the scope of alignment has expanded through online multi-agent learning (Deng & Mineiro, 2024) and self-play pipelines (Chen et al., 2024b; Singh et al., 2023), which demonstrate the capacity of LLMs to improve beyond human-labeled data. While some approaches focus on implicit reward bootstrapping (Chen et al., 2024a) or optimizing data selection within generator-verifier loops (Tu et al., 2025a), existing methods predominantly prioritize outcome-based verification. Complementing this landscape, our work bridges a critical gap by incorporating *non-verifiable* preference signals alongside standard verifiable rewards. We introduce a dual-reward mechanism that effectively leverages both signal types, enabling robust reasoning improvements even when definitive ground-truth verification is sparse or unavailable.

6. Conclusion

In this paper, we presented Dorado, a resource-efficient offline framework for reasoning. To the best of our knowledge, this is the first work to demonstrate that non-verifiable data can synergistically enhance verifiable reasoning models, effectively bridging the gap between specialized problem-solving and general conversational alignment. By integrating GC-Boost with the Dual Reward mechanism, Dorado achieves promising performance while reducing training costs by 72× compared to prominent online RL baselines. Crucially, unlike standard approaches that often sacrifice diversity for accuracy, Dorado expands the model's solution space, simultaneously boosting Pass@1 and Pass@k. Finally, as a plug-and-play solution, our framework offers a scalable and accessible path for advancing LLM reasoning capabilities under constrained computational budgets.

Impact Statements

This work aims to advance LLM reasoning by exploring post-training strategies that substantially reduce training and computational budgets. By demonstrating that strong reasoning capabilities can be achieved without expensive online optimization or large-scale training pipelines, our findings may encourage the community to reconsider existing LLM reasoning post-training practices. More broadly, lowering training budgets can help reduce barriers to entry for research on LLM reasoning, and such techniques should be developed and applied responsibly.

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A. Appendix

A.1. General Chat Boost on Other Model Series

To evaluate the broader impact of our framework, we assess the base models and their GC-Boost variants on AlpacaEval 2.0 (Dubois et al., 2025) using Qwen2-72B-Instruct as judger (qwe, 2024). As shown in Table 8, the results demonstrate that our General-Chat Boosting (GC-Boost) strategy is the pivotal factor in transforming model capabilities. It not only yields a substantial surge in general conversational proficiency but also laterally enhances the model's fundamental reasoning potential. For instance, the Qwen2.5-3B model achieves a 5× increase in Length-Controlled Win Rate (LC), rising from 6.05% to 30.72%. Similarly, the Qwen2.5-Math-1.5B model evolves from having negligible chat capabilities (0.30%) to becoming a functional assistant (7.29%) solely through GC-Boost. Notably, comparing the "+ GC-Boost" and "+ GC-Boost + DR" variants reveals that while the Dual Reward (DR) stage is essential for mathematical refinement, it contributes marginally to general chat metrics compared to the transformative impact of the GC-Boost phase. Even for the general-purpose Llama-3.2-3B-Instruct, GC-Boost further elevates its LC win rate to 39.54%. This confirms that GC-Boost is the primary driver for equipping models with robust instruction-following abilities, laying a critical foundation for advanced reasoning.

Table 8. General chat ability on AlpacaEval 2.0. We compare base models and their GC-Boost variants. LC: Length-Controlled Win Rate, WR: Win Rate. **Bold** indicates the best result within each model group.

Base Model	Variant	AlpacaEval 2.0	
		LC (%)	WR (%)
Qwen2.5-Math-1.5B	Base	0.30	0.75
	+ GC-Boost	**7.29**	**8.20**
	+ GC-Boost + DR	7.23	**8.20**
Qwen2.5-3B	Base	6.05	6.15
	+ GC-Boost	**30.72**	**31.55**
Llama-3.2-3B-Instruct	Base	21.26	22.36
	+ GC-Boost	**39.54**	**34.29**

A.2. Qualitative Evaluation Response Quality on Math Benchmarks

To assess the qualitative aspects of mathematical reasoning such as clarity and logical coherence beyond simple correctness, we employ the AlpacaEval 2.0 framework. Specifically, we utilize an LLM-as-a-judge to evaluate the quality of generated responses by computing the win rate against the reference model Qwen2.5-Math-7B. In Table 9, we benchmark our Dorado model against the offline baseline Qwen2.5-7B-DPO-VP. The results demonstrate that Dorado consistently outperforms this baseline in both Win Rate (WR) and Length-Controlled Win Rate (LC) across all datasets. This advantage is most pronounced on the challenging AIME 2024 benchmark, where Dorado achieves a dominant LC of 69.39% compared to the baseline's 41.67%, and a standard Win Rate of 70.00%. Similarly, on AMC 2023, our model secures an LC of 77.04%, surpassing the baseline's 63.70%. This indicates that our framework steers the model towards reasoning trajectories that are more accurate and better aligned with human preferences.

Table 9. Qualitative evaluation on mathematical benchmarks using the AlpacaEval 2.0 framework. We compare the response quality against the reference model Qwen2.5-Math-7B. **Bold** indicates the best performance.

Benchmark	Model	LC (%)	WR (%)
Math500	Qwen2.5-7B-DPO-VP	51.05	51.10
	Dorado	**57.09**	**55.40**
AMC 2023	Qwen2.5-7B-DPO-VP	63.70	62.50
	Dorado	**77.04**	**72.50**
AIME 2024	Qwen2.5-7B-DPO-VP	41.67	50.00
	Dorado	**69.39**	**70.00**

Table 10. Generated Token Analysis. Average token count of model responses across various mathematical benchmarks.

Model	**MATH500**	**Minerva**	**Olympiad**	**AMC23**	**AIME24**	**Avg.**
Qwen2.5-Math-7B	651	577	695	662	748	**667**
Qwen2.5-Math-7B-Instruct	641	650	886	911	1164	850
Eurus-2-7B-PRIME	655	822	897	1020	1164	912
Qwen2.5-7B-Simple-RL	588	775	801	767	952	777
Qwen2.5-7B-PURE-VR	626	646	863	850	1050	807
Qwen2.5-7B-DPO-VP	654	691	924	886	1251	881
Qwen2.5-Math-7B-OpenR1	2440	3125	3351	3116	3892	3185
Dorado	587	553	797	765	979	736